

Bhavishya Chebrolu

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ABOUT ME

Undergraduate researcher working at the intersection of Machine Learning and High Energy Physics, with experience in density estimation and likelihood-ratio methods for dark matter searches and unsupervised anomaly detection using CMS open data. Co-author of three arXiv preprints on machine-learning-based searches with CMS open data.

EDUCATION

Mohan Babu University, Tirupati, India

2023 – 2027

B.Tech

CGPA: 9.19 / 10

Sree Vidyanikethan International School, Tirupati, India

2023

CBSE (10+2)

Grade: 92.4 / 100

SELECTED PROJECTS

The Unlikely Lab — Research Group Website

Next.js, TypeScript, MDX, Tailwind CSS

Site: the-unlikely-lab.github.io/the-unlikely-lab • Code: github.com/The-Unlikely-Lab/the-unlikely-lab

Mono-Z Analysis

Python, PyTorch, uproot, MadGraph5

Leptonic channels: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

Hadronic channel: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT-

- Co-developed the public analysis code for both mono-Z papers, maintained as two channel-specific repositories. Leptonic ($\mu\mu / ee$): CMS MiniAOD extraction with uproot, cleaning, EDA, Neural Spline Flow training/scoring, and profile-likelihood CLs fits. Hadronic ($Z \rightarrow jj$, CMS Run 2015D HTMHT sample): event selection, conditional flow-matching background model, and projected CLs sensitivity for vector and axial-vector mediators.

RESEARCH EXPERIENCE

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows

CMS Run 2015D Open Data

B. Chebrolu, H. Rasineni. arXiv preprint, 2026 (submitted for journal publication).

DOI: 10.48550/arXiv.2607.13771

- Co-developed a mono-Z dark matter search on 2.32 fb^{-1} of CMS Run 2015D data (parallel $\mu\mu/ee$ channels) using Neural Spline Flows: channel-specific SM background flows and mediator-specific signal flows, with the per-event log-likelihood ratio $\log p(x|\text{DM}) - \log p(x|\text{SM})$ as the test statistic followed by a simultaneous SR+VR binned profile-likelihood fit to set 95% CL limits on the signal strength.

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching

CMS Run 2015D HTMHT Open Data

B. Chebrolu, H. Rasineni. arXiv preprint, 2026.

DOI: 10.48550/arXiv.2609.02923

- Co-developed the full simulation-to-inference pipeline (MadGraph5_aMC@NLO $\rightarrow$ Pythia 8 $\rightarrow$ Delphes signal generation) with a conditional flow-matching normalizing flow trained on real HTMHT events, including sentinel imputation for missing-object features and projected expected significances of 2.89σ , 7.62σ , and 7.41σ for three simplified-model benchmarks.

Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events

CMS Open Data Dijet Sample

B. Chebrolu, H. Rasineni, P.A. Immadi. arXiv preprint, 2026.

DOI: 10.48550/arXiv.2609.06686

- Co-developed an unsupervised search for anomalous dijet events using neural spline flow density estimation: a normalizing flow trained on jet, dijet, and event-level observables to learn the Standard Model background directly from data without a signal hypothesis, flagging low-likelihood events, with extensive validation via mass-decorrelation tests, permutation-based null tests, and training-stability studies.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript / TypeScript (React, Next.js)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, Density Estimation (Normalizing Flows, Likelihood Ratios)

Web/Platform: Next.js, React, TypeScript, Tailwind CSS, MDX

Tools/Infra: Git, GitHub Actions (CI/CD), Docker, Pandoc, L^AT_EX, uproot, Google Colab

CERTIFICATIONS

- **Object Tracking and Motion Detection with Computer Vision** — MathWorks (Coursera), 2026 (Credential)
- **Machine Learning for Computer Vision** — MathWorks (Coursera), 2026 (Credential)
- **Bayesian Statistics: Mixture Models** — UC Santa Cruz (Coursera), 2026 (Credential)
- **Pattern Discovery in Data Mining** — University of Illinois Urbana-Champaign (Coursera), 2026 (Credential)
- **Web Development Internship (HTML, CSS, JavaScript)** — ApexPlanet Software Pvt Ltd, May–July 2026 (Certificate ID: APSPL2646294)